Certified Tsirelson bounds for 229 tight bipartite Bell inequalities

Claude Opus 5.5 Extra Effort[1]

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The maximal quantum violations of the tight bipartite Bell inequalities with up to five binary settings per party, catalogued by Brunner and Gisin, Pál and Vértesi, and Ito, Imai and Avis, are known only from floating-point computations. We give certified windows for 229 of these 239 inequalities. Each window consists of an exact quantum strategy, whose violation is a rational number, and an exact sum-of-squares certificate in the algebra of projectors. A rigorously bounded diagonal shift makes the certificate valid for all commuting-operator strategies. 223 windows are narrower than 10^{-7} and 146 are narrower than 10^{-8}. For the inequalities that Pál and Vértesi left open, J^{97}_{4422} is now closed to within 2.2×10^{-8} of its two-qubit value. I^{20}_{4422}, J^{34}_{4422} and A_{65}, previously closed only numerically, are closed rigorously. The windows of J^{73}_{4422}, J^{62}_{4422}, J^{30}_{4422} and J^{48}_{4422} shrink by factors of 10, 11, 6 and 3. Real strategies in local dimension 26–31 raise the best lower bounds of A_{68}, A_{89}, A_{62} and A_{82}. The best strategies for J^{73}_{4422} and J^{62}_{4422} form ladders of nearly degenerate Schmidt pairs, and their violations converge geometrically with the number of pairs. As for I_{3322}, the maximum appears to be approached only in infinite dimension. We also find that the published maximum of A_{19} is 4.4×10^{-7} too low. The candidates were found by a multi-agent language-model system working against a frozen exact verifier, and all certificates are released with this work.

I. INTRODUCTION

A Bell inequality bounds the correlations of local hidden-variable models [1, 2]. Its quantum maximum, the Tsirelson bound [3], measures how far quantum mechanics exceeds that limit. Tsirelson bounds are needed for device-independent protocols, dimension witnesses and self-testing [4]. Computing them is hard in general. Lower bounds come from explicit strategies, found by optimization over states and measurements in a fixed dimension [5, 6]. Upper bounds come from the Navascués–Pironio–Acín (NPA) hierarchy of semidefinite relaxations [7–9]. Both sides are usually computed in floating point. An SDP solver's output is not a proof: its dual can be slightly infeasible, and its reported bound can lie below the true maximum. Naceur, Wang, Magron and Acín recently showed how often this happens. For the 86 inequalities of the Ito–Imai–Avis list that need relaxation order 2 or higher, 32 floating-point NPA bounds lay below the true maximum. They also showed how to turn solver output into exact rational upper bounds [10].

Here we certify both sides for a whole catalog. Brunner and Gisin listed 26 tight inequalities with four binary settings per party (the 4422 scenario) [11]; the complete list has 175 classes [17]. Pál and Vértesi added 129 more (their J list) and computed lower and upper bounds for all 241 tight inequalities then known with up to five settings [12]. For all but 20 the two bounds met numerically. In a follow-up they studied I_{3322} in infinite dimension, raised eleven lower bounds with an iterative method in local dimension up to 25, and closed four more cases numerically [13]. The numerical windows that remain are listed in Table I. I_{3322} itself has since been settled rigorously: its supremum is not attained in finite dimension [14–16], and its value is known to 55 digits [18].

We report three results. First, a certified table: 229 of the 239 inequalities in our catalog have an exact strategy and an exact upper-bound certificate, and 223 of these windows are narrower than 10^{-7} (Table II, Appendix A). Second, progress on the open cases: one closed, three closed rigorously that were closed only numerically, four narrowed by factors 3–11, and four lower bounds raised (Table I). Third, structure. The best strategies for the hardest cases are ladders of Schmidt pairs whose value converges geometrically with dimension (Fig. 1), and one published "tight" value turns out to be wrong (Sec. V D).

The work was done by a multi-agent system of language-model agents that searched for candidates and submitted them to a frozen verifier. The verifier was written and tested before any search began and revised once during the first run. Only its verdicts count, and every final result was checked again by independent code (Sec. VI).

II. SETTING

Alice measures one of m_A two-outcome observables and Bob one of m_B. In the Collins–Gisin form used by Pál and Vértesi [12, 19], an inequality reads

$$\sum_x b^A_x\, p(A_x)+\sum_y b^B_y\, p(B_y)+\sum_{x,y} b^{AB}_{xy}\, p(A_xB_y) \le b_0, \quad (1)$$

where $p(A_x)$ is the probability that Alice's setting x gives outcome 1, and similarly for Bob and for the joint outcome. In a quantum strategy, $p(A_x) = \langle\psi|A_x\otimes\mathbb{1}|\psi\rangle$ and $p(A_xB_y) = \langle\psi|A_x\otimes B_y|\psi\rangle$, with orthogonal projectors A_x, B_y. Projective measurements and pure states suffice [20]. We write W for the Bell operator and call $\langle\psi|W|\psi\rangle - b_0$ the violation. All quantities below are violations. Every inequality in our catalog has local bound $b_0 = 0$ in the form used, except I^7_{4422} ($b_0 = 1$).

The catalog has 239 inequalities: CHSH; I_{3322}; the 20 Brunner–Gisin inequalities I^k_{4422} together with A_5,

A_6, $\mathrm{AS}_{1,2}$ and $\mathrm{AII}_{1,2}$ from the same appendix [11]; the 129 Pál–Vértesi inequalities J^k_{4422} [12]; and 82 further inequalities of the Ito–Imai–Avis A list [21, 22], taken from the repository of Ref. [10]. This is Pál and Vértesi's set of 241 inequalities without the three 3422 inequalities of Collins and Gisin [19], plus CHSH. We checked every entry exactly. Its local bound, computed by enumerating the deterministic strategies, equals b_0, and the deterministic points that saturate it span a facet. One repository file (A_{10}) uses the opposite sign convention from the rest, and this test caught it. As a check of the coefficients, see-saw optimization reproduces the published violation to 10^{-7} for all A-list inequalities whose published optimum lies in local dimension at most 3, and for A_{10} in dimension 5. For the 4422 inequalities the certified windows of Sec. V provide the same check.

Our reference values are the windows of Pál and Vértesi [12, 13], printed to seven decimals. For the open A-list cases, the repository of Ref. [10] also lists numerical NPA values at relaxation order 4. They are lower than the 2010 upper bounds, and we use them as the upper reference although they are not certified.

III. CERTIFICATES

A. Lower bounds

A lower-bound certificate is a strategy with exact entries. Each projector is given by linearly independent spanning vectors with rational or Gaussian-rational entries, collected as the columns of V. The verifier forms $P = V(V^\dagger V)^{-1}V^\dagger$ in exact arithmetic, so P is an orthogonal projector by construction. It also checks $P = P^\dagger = P^2$. With an exact state vector ψ, the violation $\langle\psi|W|\psi\rangle/\langle\psi|\psi\rangle - b_0$ is a rational number. It is computed twice. The first computation uses the reduced-state formula $\langle\psi|P\otimes Q|\psi\rangle = \mathrm{Tr}(P\,R\,Q^T R^\dagger)$, where $R_{ab} = \psi_{ad_B+b}$. The second uses an entrywise Kronecker sum (for $d_A d_B \le 64$) or the formula reduced on Bob's side instead, and the two results must agree. Strategies found numerically are rationalized to denominators of order 10^{12}. The projector is rebuilt exactly from the rounded vectors, so the exact value is a valid lower bound whatever the rounding. Near a stationary point the rounding changes the value only at second order.

B. Upper bounds

Let $\mathcal{P}$ be the universal $*$-algebra generated by self-adjoint idempotents $p_1,\dots,p_{m_A}$ and $q_1,\dots,q_{m_B}$ with $p_x q_y = q_y p_x$. The reduced words (α,β), where α is a string over Alice's letters and β one over Bob's, both without equal neighbors, form a linear basis of $\mathcal{P}$. An upper-bound certificate consists of words $w_1,\dots,w_N$, a rational symmetric matrix Y and the identity

$$\sum_{i,j=1}^{N} Y_{ij}\, w_i^* w_j = \lambda\mathbb{1} - W \quad \text{in } \mathcal{P}. \tag{2}$$

The verifier reduces every product $w_i^* w_j$ to normal form and checks Eq. (2) coefficient by coefficient in integer arithmetic. The value of λ is read off as the coefficient of $\mathbb{1}$. If $Y \succeq 0$, Eq. (2) is a sum of squares and λ is an upper bound. This is the dual of the NPA relaxation with moment basis $\{w_i\}$ [8, 9, 23]. A rational Y obtained by rounding a numerical dual is rarely positive semidefinite, however, because the optimal dual is singular. Instead of insisting on exactness [10, 24] we pay for the defect.

Theorem 1. *Suppose Eq. (2) holds and $Y + \tau\mathbb{1} \succeq 0$ for some $\tau \ge 0$. Then every commuting-operator strategy (projectors A_x, B_y with $[A_x, B_y] = 0$ on any Hilbert space, unit vector ψ) satisfies $\langle\psi|W|\psi\rangle \le \lambda + N\tau$. In particular this holds for every finite-dimensional tensor-product strategy.*

Proof. The strategy defines a $*$-representation of $\mathcal{P}$. Let $\Gamma_{ij} = \langle w_i\psi, w_j\psi\rangle$. Then Γ is Hermitian and positive semidefinite, and $\Gamma_{ii} = \|w_i\psi\|^2 \le 1$ because each w_i is a product of projectors. Taking the expectation of Eq. (2) gives $\lambda - \langle\psi|W|\psi\rangle = \sum_{ij} Y_{ij}\Gamma_{ij} = \mathrm{Tr}[(Y+\tau\mathbb{1})\bar\Gamma] - \tau\,\mathrm{Tr}\,\Gamma \ge -\tau N$, since $\bar\Gamma \succeq 0$. □

The verifier finds a small τ with a rigorous proof. For a shift σ it computes a floating-point Cholesky factor L of $Y + \sigma\mathbb{1}$. It then encloses the residual $F = Y + \sigma\mathbb{1} - LL^T$ entrywise by interval arithmetic, rounding each operation to nearest and widening it by one unit in the last place in each direction. Then $\tau = \sigma + f$ with $f \ge \|F\|_F$ works, since $Y + \tau\mathbb{1} = LL^T + (F + f\mathbb{1}) \succeq 0$. This is the residual argument used in verified positive-definiteness tests [25]. An earlier version used interval Cholesky factorization. It overestimated badly for the nearly singular matrices that optimal duals produce: it needed $\tau \sim 10^{-5}$ where 10^{-10} suffices. The verifier searches σ on a geometric ladder and reports the certified bound $\lambda + N\tau - b_0$. For $N \le 120$ it also proves $Y + 2\tau\mathbb{1} \succ 0$ exactly by fraction-free (Bareiss) elimination [26]. For the 229 final upper certificates the penalty $N\tau$ has median 3×10^{-9} and maximum 9×10^{-8}. Because the certificate is a sum of squares in $\mathcal{P}$, the bound holds for the commuting-operator value, so each window brackets both that value and the tensor-product supremum.

C. Verification protocol

The verifier is about 800 lines of Python using only the standard library, plus a 77-line C program for the residual bound. Its sources, the inequality catalog and the reference values are frozen and hash-pinned. A submission is frozen before it is checked, and the verifier, not the submitter, records the verdict. Its tests reproduce the

published values, exercise every verdict path, reject malformed input, and compare against independent implementations on random instances: a separate Bell-value evaluation, a string-rewriting implementation of the word algebra, and the shift search applied to matrices of exactly known spectrum, built from Cayley-rational orthogonal matrices. Every result was re-checked with separately written code. Strategies were evaluated in exact Gaussian-rational arithmetic. For upper certificates the re-check used its own word reduction, an exact identity check, and an exact proof of positive definiteness, either by Bareiss elimination (all windows in Table I) or by an exact-integer residual bound (most of the catalog). All 467 certificates behind the table in Appendix A passed. A consistency monitor confirmed that no certified lower bound exceeds a certified upper bound for any inequality.

IV. FINDING THE CERTIFICATES

A. Strategies

Seesaw optimization alternates three steps [5, 6]: the top eigenvector of W, and for each projector the positive spectral part of its effective operator. From random starts it mostly finds qubit optima, as Pál and Vértesi noted [12]. Their 2010 method keeps the state in Schmidt form $\sum_i \lambda_i |ii\rangle$, so each step involves only $n \times n$ matrices [13]. Two extensions produced our new lower bounds.

Bulk-cell insertion. The best strategies for J_{4422}^{73} and J_{4422}^{62} have Schmidt coefficients in nearly degenerate pairs that decay geometrically, and projectors that couple neighboring pairs, like an open chain. Basin hopping in local dimension 12–16 finds a short chain; for J_{4422}^{73} it has six pairs and violation 1.1217770. In the Schmidt basis we then duplicate one bulk pair together with its coupling block, add a small random perturbation, and re-optimize. This adds one pair (or several) at a time up to the verifier's limit of local dimension 32 (16 pairs). For J_{4422}^{62} the dimension-32 strategy came from inserting several pairs into an eight-pair chain at once.

Ancilla doubling. For the five-setting A-list inequalities we embed an optimum as $A_x \otimes \mathbb{1}_2$, $B_y \otimes \mathbb{1}_2$, $\psi \otimes |\Phi_2\rangle$, which keeps the value. A random perturbation breaks the twofold degeneracy, and re-optimizing climbs to a new branch. We also doubled only the K Schmidt levels of smallest weight, and compressed each strategy to its active rank. Repeating these steps from local dimension 5–6 gave the A_{68}, A_{89}, A_{62} and A_{82} strategies in dimension 26–31. The same procedure reproduces Pál and Vértesi's $\mathbb{R}^{17}$ optimum for J_{4422}^{30} (0.4363842240) from a real 7-dimensional optimum. It saturates there: partial doublings and complex deformations found nothing higher.

B. Dual certificates

Level 3 of the hierarchy for four settings per party has 217 words in our basis, and level 4 has 865. For five settings the counts are larger still. Off-the-shelf first-order solvers were too inaccurate for certificates, and interior-point solvers needed too much memory on the available machine. The agents therefore wrote a primal–dual interior-point method with the HKM search direction [27–29]. It forms the Schur complement panel by panel in half-vectorized storage and uses the sparsity of each constraint, whose matrix has only a few nonzero rows. Every Schur solve is iteratively refined against the exact operator, which keeps the primal infeasibility near 10^{-10}. Moment matrices with $N \approx 450$ words needed 2.6–2.9 GB and 15–40 minutes per warm-started round.

Words beyond level 3 were chosen mainly by column generation. Given the primal moment matrix Γ of the current optimum, a candidate word u is scored by

$$s(u) = d - c_K^T \, (\Gamma_{KK} + \epsilon \mathbb{1})^{-1} c_K, \tag{3}$$

where K indexes the current words w_i for which $w_i^* u$ reduces to a known moment, c_i is that moment, and d is the moment of $u^* u$. A negative score means the current moment matrix cannot be extended by a row for u, so adding u cuts off the current optimum. Each round added the 10–60 most negative words of degree up to 4 and dropped words of low weight. Simpler heuristics based on the diagonal of the moment matrix supplemented the pricing. Scores were computed at $\epsilon = 10^{-8}$ to 10^{-9}; at 10^{-6} no candidate had a negative score. The dual was rounded to a dyadic denominator (2^{46} or 2^{47}). The residual of each moment class in Eq. (2) was spread evenly over the class's entries or placed on a single entry.

For the catalog, a batch pipeline computed a seesaw strategy at the published dimension and an NPA dual at the published level, with Clarabel [30] for $N \lesssim 100$ and the custom solver above that. It raised the level until the window closed.

V. RESULTS

A. The catalog

Table II summarizes the certified table. All 129 inequalities of the J list and all 20 of the Brunner–Gisin list are certified on both sides, with windows below 10^{-7} except for the four open J cases. Of the 238 certified lower bounds, 193 come from qubit strategies (148 real, 45 complex), 34 from local dimension 3–5, and 11 from local dimension 8–32. The upper certificates use between 9 and 454 words (median 97). The closed windows have median width 4.5×10^{-9} and maximum 8.7×10^{-8}. Ten inequalities lack a two-sided certificate. They are I_{3322}, which is settled elsewhere [18]; eight open A-list cases, for which no upper certificate was computed (full level 3 for

TABLE I. Windows on the maximal quantum violation for the inequalities that Pál and Vértesi left open [12, 13]. Literature: lower bound and upper bound (lower bound plus the gap at their deepest NPA level) from Ref. [13], Table I, printed to 10^{-7}. ‡Upper bound from the numerical NPA order-4 values in the repository of Ref. [10] (not certified). Certified: exact strategy and exact sum-of-squares certificate verified as in Sec. III. Values are rounded outward to 10^{-10}. *Improves on the literature value. †Literature value, not certified here. §Closed numerically in Ref. [13] (for I^{20}_{4422} and J^{34}_{4422} at level 3 plus a quarter of the $aa'bb'$ words). A dash means that no certified upper bound reaches the literature value.

	literature		this work		width	
inequality	lower	upper	lower	upper	literature	this work
J^{97}_{4422}	1.1583626	1.1585051	1.1583625533	1.1583625755*	1.4×10^{-4}	2.2×10^{-8}
I^{20}_{4422}	0.4676794	0.4676794	0.4676793861	0.4676793911	0§	5.0×10^{-9}
J^{34}_{4422}	0.4492657	0.4492657	0.4492657413	0.4492657466	0§	5.3×10^{-9}
A_{65}	0.3688996	0.3688996	0.3688995748	0.3688995760	0§	1.2×10^{-9}
J^{73}_{4422}	1.1223170	1.1225579	1.1223870369*	1.1224104368*	2.4×10^{-4}	2.3×10^{-5}
J^{62}_{4422}	0.7500755	0.7500921	0.7500885544*	0.7500900835*	1.7×10^{-5}	1.5×10^{-6}
J^{30}_{4422}	0.4363842	0.4372629	0.4363842240	0.4365201856*	8.8×10^{-4}	1.4×10^{-4}
J^{48}_{4422}	0.7516220	0.7516340	0.7516219860	0.7516256937*	1.2×10^{-5}	3.7×10^{-6}
A_{68}	0.4025522	0.4050294‡	0.4028002584*	0.4050294†	2.5×10^{-3}	2.2×10^{-3}
A_{89}	0.3035637	0.3054510‡	0.3039223061*	0.3054510†	1.9×10^{-3}	1.5×10^{-3}
A_{62}	0.4065268	0.4066728‡	0.4065717137*	0.4066728†	1.5×10^{-4}	1.0×10^{-4}
A_{82}	0.4708838	0.4709172‡	0.4709112502*	0.4709172†	3.3×10^{-5}	5.9×10^{-6}
A_{84}	0.6352087	0.6352107‡	0.6352087†	0.6352107†	2.0×10^{-6}	2.0×10^{-6}
A_{21}	0.3260601	0.3260654‡	0.3260600674	0.3260654†	5.3×10^{-6}	5.3×10^{-6}
A_{14}	0.4759513	0.4768133‡	0.4759513†	0.4768133†	8.6×10^{-4}	8.6×10^{-4}
A_{47}	0.4608544	0.4615622‡	0.4608543948	0.4615622†	7.1×10^{-4}	7.1×10^{-4}
A_{64}	0.3900890	0.3905074‡	0.3900890509	0.3905074†	4.2×10^{-4}	4.2×10^{-4}
A_{67}	0.3990671	0.3990671	0.3990671029	0.3990671†	0§	–
A_{80}	0.3769863	0.3769863‡	0.3769863185	0.3769863†	0‡	–

TABLE II. Certified windows across the catalog. Columns: number of inequalities, number certified on both sides, and number with certified window below 10^{-7} and below 10^{-8}. A_5 and A_6, which also appear in the Brunner–Gisin appendix, are counted in the A-list row. The full table is in Appendix A.

family	all	both	$< 10^{-7}$	$< 10^{-8}$
Pál–Vértesi J list (4422)	129	129	125	68
Brunner–Gisin I_{4422} list	20	20	20	12
Ito–Imai–Avis A list (up to 5522)	84	75	73	63
CHSH, I_{3322}, $AS_{1,2}$, $AII_{1,2}$	6	5	5	3
total	239	229	223	146

these five-setting inequalities has 436 words, beyond our memory budget); and A_{80}, which the numerical values of Ref. [10] already close. Two further A-list inequalities (A_{21}, A_{67}) have upper certificates that are not yet tight and remain wider than 10^{-7}.

The certified windows also test the published values. For every closed inequality except A_{19}, the printed value lies within 5×10^{-8} above or 10^{-7} below our window. In eight cases (J^{63}_{4422}, J^{81}_{4422}, J^{103}_{4422}, J^{106}_{4422}, J^{124}_{4422}, A_{38}, A_{52}, A_{81}) it lies 5–7 $\times 10^{-8}$ below the window, more than rounding to seven decimals allows. Truncation, or a slightly suboptimal strategy, would explain this. A_{19} is discussed in Sec. V D.

B. Open cases

Table I compares the literature windows with ours.

J^{97}_{4422}. Pál and Vértesi found a two-qubit strategy with violation 1.1583626 and an upper bound 1.4×10^{-4} higher at level 3. They reported that a level-3-plus-$aa'bb'$ computation gave a value below their lower bound, which signals inaccuracy, and concluded that the qubit strategy is very probably optimal [13]. Our certificate with 300 words (full level 3 plus 83 selected words) proves ≤ 1.1583625755. An exact real-qubit strategy reaches 1.1583625533, so the window has width 2.2×10^{-8}. (The printed lower bound 1.1583626 is rounded up.) The raw sum-of-squares value is $\lambda = 1.1583625531$, just below the strategy value, so the whole width comes from the penalty $N\tau$. Proving exact optimality of the qubit value would need an exact certificate.

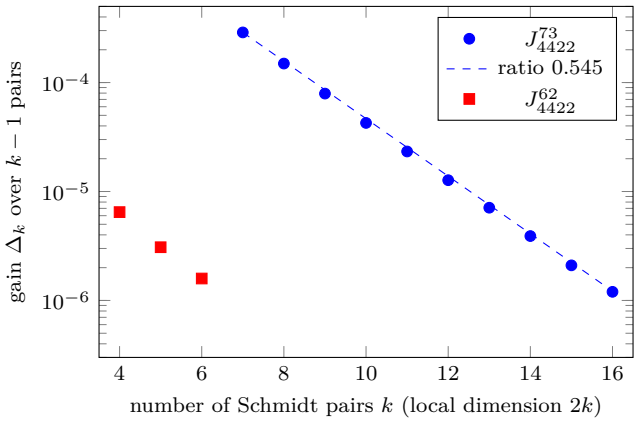


FIG. 1. Ladder strategies. Gain in violation from adding one Schmidt pair by bulk-cell insertion, for J^{73}_{4422} (from six pairs, 1.1217770, to sixteen pairs, 1.1223870) and J^{62}_{4422} (from Pál and Vértesi's three-pair $\mathbb{R}^6$ optimum). The gains decrease geometrically. Extrapolating the J^{73}_{4422} series gives a limit ≈ 1.1223884, 2.2×10^{-5} below the certified upper bound.

I^{20}_{4422}, J^{34}_{4422} and A_{65}. These were closed numerically in 2010. Exact strategies in local dimension 4 and certificates with 277 words close the first two to 5×10^{-9}. A strategy in dimension 5 and a certificate with 276 words close A_{65} to 1.2×10^{-9}. Our certificates for A_{67} and A_{80}, also closed numerically, are not yet tight.

J^{73}_{4422} and J^{62}_{4422}. Both sides move. The lower bounds are the ladders of Fig. 1. Already at local dimension 20 the J^{73}_{4422} ladder gives 1.1223367, above Pál and Vértesi's 1.1223170 in the same dimension. At dimension 32 it gives 1.1223870369. The upper bounds come from column generation with 420 and 311 words. The windows shrink from 2.4×10^{-4} to 2.3×10^{-5} and from 1.7×10^{-5} to 1.5×10^{-6}.

J^{30}_{4422} and J^{48}_{4422}. The upper bounds fall by 7.4×10^{-4} and 8.3×10^{-6}, with 454 and 442 words. The lower bounds are Pál and Vértesi's real optima in dimensions 17 and 12, re-derived and certified exactly. Pál and Vértesi expected the J^{30}_{4422} lower bound not to be optimal [13]. Our ladder constructions saturate at their value. The window for J^{30}_{4422} is therefore still 1.4×10^{-4} wide.

A list. Ancilla doubling raises the lower bounds of A_{68} (by 2.5×10^{-4}, real dimension 26), A_{89} (3.6×10^{-4}, dimension 31), A_{62} (4.5×10^{-5}, dimension 27) and A_{82} (2.7×10^{-5}, dimension 26). For A_{82} this leaves 5.9×10^{-6} to the numerical order-4 value. On the upper side our certificates did not reach the order-4 values of Ref. [10]. A rigorous five-setting relaxation beyond level 4 remains to be done.

C. Ladders and infinite dimension

The ladder values in Fig. 1 grow monotonically with dimension, and their increments decay with a nearly constant ratio, 0.52–0.59 for J^{73}_{4422} (least-squares fit 0.545). The pattern matches the construction of Pál and Vértesi for I_{3322} [13], whose supremum is now known not to be attained in finite dimension [14, 15]. If no other family of strategies does better, the maximal violations of J^{73}_{4422} and J^{62}_{4422} are approached only as the local dimension grows without bound. The extrapolated limits are ≈ 1.1223884 for J^{73}_{4422} and about 0.7500886 for J^{62}_{4422} (its dimension-32 value is 0.7500885544). These lie 2.2×10^{-5} and 1.5×10^{-6} below the certified upper bounds. So either the hierarchy converges slowly for these inequalities, or a different family of strategies does better. For J^{30}_{4422} and J^{48}_{4422} the analogous constructions saturate at Pál and Vértesi's values.

D. A published value that is too low

For A_{19}, Pál and Vértesi report violation 0.6226300 from a complex qubit strategy, with their level-2a NPA bound equal to it [12]. The same value appears as the exact maximum in the benchmark data of Ref. [10]. A real strategy in local dimension 4 with rank-2 projectors reaches 0.6226304433, and a certificate with 318 words bounds the maximum by 0.6226304480. The true maximum is therefore 4.4×10^{-7} above the published value, beyond rounding or truncation. Since every NPA level bounds the maximum from above, the published NPA value must be inaccurate. The printed strategy value matches the best complex-qubit strategy, 0.6226299743, a local optimum. This is a small but concrete example of why floating-point "tight" values need certificates.

VI. HOW THE RESULTS WERE FOUND

The work used two four-hour runs of a multi-agent system of language-model agents (Claude Sonnet 5.5, Anthropic), run as non-interactive coding sessions in 40-minute shifts on one laptop with ten cores and 16 GB of memory. A supervising session (Claude Opus 5.5) built the verifier, the catalog and the reference values before the first run. It also coordinated the agents and audited their results. The agents shared a message board and could not modify the verifier, the data or the board software. The board recorded 550 verified submissions. 481 came from the catalog certification and 64 improved on the reference values in force at the time. Two of these were revoked when the 2010 table superseded the 2009 one. The other five, by the auditing agent, re-derived published optima or were calibration runs.

First run. Two agents searched for strategies, one for dual certificates, and one audited. Three minutes after launch, a complex strategy in local dimension 5 beat the 2009 value for J^{30}_{4422}. Within seven minutes the auditor found Ref. [13], which had already superseded the 2009 windows. The reference values were rebuilt from the 2010 table, the verifier was re-frozen, and two early wins were revoked as superseded. Later in the run the verifier was

revised once: the cap on the local dimension was raised from 16 to 32, and a faster second evaluation was added for large strategies. It was re-tested and re-frozen. The certificate agent found that a first-order solver could not deliver certificate accuracy and wrote the interior-point method of Sec. IV. Its first improvement (J^{97}_{4422}) came 38 minutes after launch, and the closure of J^{97}_{4422} after 3 hours 11 minutes. A strategy agent found the ladder structure of J^{62}_{4422} and J^{73}_{4422} within 75 minutes.

Second run. The reference values were raised to the certified results of the first run. Five agents took part: two for dual certificates, one for strategies, one for the catalog certification, and the auditor, who began with a citation search. That search found Ref. [10], whose repository supplied the A-list coefficients. The supervising session re-fetched and parsed them independently, checked them against the published values, and added them to the frozen catalog 38 minutes after launch. The first raises of the four A-list lower bounds followed within two hours, by ancilla doubling, and the values in Table I within four.

Each mid-run change of reference values followed a literature finding by the auditing agent. The auditor also re-checked every improvement independently, spot-checked the catalog certificates, and red-teamed the verifier with several hundred random and adversarial inputs. None was wrongly accepted.

VII. DISCUSSION

Certified windows turn a table of numerical estimates into a table of theorems. For most of the catalog the maximal violation is now known rigorously to eight or nine digits. The method rests on two ingredients that are cheap to trust: exact rational linear algebra, and one rigorous bound on the smallest eigenvalue of an explicit matrix (Theorem 1). The cost is a penalty $N\tau$, typically 10^{-9}–10^{-8} and at most 9×10^{-8} here. Unlike exact rounding schemes [10, 24], it needs no facial reduction and does not require the rounded matrix to remain positive semidefinite, so it works directly with solver output.

Three limitations remain. The windows are not exact values. Proving that a maximum equals an algebraic number, for example the qubit value of J^{97}_{4422}, would need exact certificates over a number field. The five-setting A-list cases need full level 3 (436 words) and beyond, which exceeded the memory available to our dense interior-point method. Symmetry reduction or term sparsity [10] are natural next steps. Finally, our literature search used arXiv and general web search and could not access citation databases, so a complete citation check of Refs. [12, 13] is still needed.

Several questions are open. Do J^{73}_{4422} and J^{62}_{4422} require infinite dimension? A proof along the lines of Refs. [14, 15] might be within reach. Is the lower bound of J^{30}_{4422} optimal, contrary to Pál and Vértesi's expectation? And which of the A-list windows close at order 5?

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DATA AVAILABILITY

All certificates (strategies and SOS certificates, as exact JSON files), the verifier, its tests, the independent checkers, the inequality catalog with its sources, and the search programs are in the project repository. The certified table of Appendix A is generated from the certificates by a script in the repository.

Appendix A: Certified table

For each inequality we list the published maximal violation (Pál and Vértesi; for the open cases the 2010 lower bound) and the best certified lower and upper bounds, rounded outward to 10^{-10}. Widths are computed from these rounded values.

TABLE III: Certified windows for all 239 inequalities of the catalog. Names: Jk = J^k_{4422} (Pál–Vértesi), Ik = I^k_{4422} (Brunner–Gisin), Ak = Ito–Imai–Avis; "pub." is the published maximal violation (for open cases the 2010 lower bound; for CHSH, Tsirelson's value).

ineq.	pub.	lower	upper	width	ineq.	pub.	lower	upper	width
CHSH	0.2071068	0.2071067811	0.2071067813	2.0×10^{-10}	J99	0.8990726	0.8990725717	0.8990725901	1.8×10^{-8}
I3322	0.2508754	–	–	–	J100	1.2675376	1.2675376399	1.2675376458	5.9×10^{-9}
I1	0.2878683	0.2878683045	0.2878683229	1.8×10^{-8}	J101	1.0296105	1.0296104781	1.0296104908	1.3×10^{-8}
I2	0.6213712	0.6213711737	0.6213711757	2.0×10^{-9}	J102	0.6650865	0.6650865119	0.6650865360	2.4×10^{-8}
I3	0.4142136	0.4142135623	0.4142135644	2.1×10^{-9}	J103	1.0520136	1.0520136544	1.0520136560	1.6×10^{-9}
I4	0.4142136	0.4142135623	0.4142135670	4.7×10^{-9}	J104	1.5875885	1.5875885171	1.5875885447	2.8×10^{-8}
I5	0.4364917	0.4364916731	0.4364916737	6.0×10^{-10}	J105	1.0742228	1.0742228124	1.0742228126	2.0×10^{-10}
I6	0.4494897	0.4494897427	0.4494897472	4.5×10^{-9}	J106	0.8382498	0.8382498540	0.8382498625	8.5×10^{-9}
I7	0.4548373	0.4548373063	0.4548373168	1.1×10^{-8}	J107	0.9339703	0.9339702504	0.9339702628	1.2×10^{-8}
I8	0.4877681	0.4877681239	0.4877681290	5.1×10^{-9}	J108	0.9676411	0.9676410648	0.9676410678	3.0×10^{-9}
I9	0.4616842	0.4616841752	0.4616842329	5.8×10^{-8}	J109	1.7260512	1.7260512223	1.7260512230	7.0×10^{-10}
I10	0.6139456	0.6139455921	0.6139455965	4.4×10^{-9}	J110	0.9457465	0.9457465079	0.9457465106	2.7×10^{-9}

ineq.	pub.	lower	upper	width	ineq.	pub.	lower	upper	width
I11	0.6383543	0.6383543206	0.6383543213	7.0×10^{-10}	J111	0.7575890	0.7575890486	0.7575890518	3.2×10^{-9}
I12	0.6188142	0.6188142246	0.6188142346	1.0×10^{-8}	J112	0.6247585	0.6247585341	0.6247585423	8.2×10^{-9}
I13	0.4348553	0.4348553293	0.4348553636	3.4×10^{-8}	J113	0.8483764	0.8483764114	0.8483764371	2.6×10^{-8}
I14	0.4794103	0.4794102632	0.4794102730	9.8×10^{-9}	J114	1.0653155	1.0653155114	1.0653155129	1.5×10^{-9}
I15	0.4348553	0.4348553293	0.4348553420	1.3×10^{-8}	J115	1.0098044	1.0098043801	1.0098043909	1.1×10^{-8}
I16	0.4142136	0.4142135623	0.4142135625	2.0×10^{-10}	J116	0.7651941	0.7651940887	0.7651941373	4.9×10^{-8}
I17	0.6714085	0.6714085444	0.6714085482	3.8×10^{-9}	J117	0.9721185	0.9721184718	0.9721184749	3.1×10^{-9}
I18	0.6429670	0.6429670025	0.6429670214	1.9×10^{-8}	J118	0.5283248	0.5283247658	0.5283247675	1.7×10^{-9}
I19	0.4971707	0.4971706561	0.4971707174	6.1×10^{-8}	J119	0.8986088	0.8986088257	0.8986088263	6.0×10^{-10}
I20	0.4676794	0.4676793861	0.4676793911	5.0×10^{-9}	J120	0.9333181	0.9333180866	0.9333180894	2.8×10^{-9}
J1	0.4554438	0.4554438200	0.4554438222	2.2×10^{-9}	J121	0.5970802	0.5970801934	0.5970801939	5.0×10^{-10}
J2	0.6140029	0.6140029009	0.6140029051	4.2×10^{-9}	J122	0.7931031	0.7931030684	0.7931030740	5.6×10^{-9}
J3	0.8106306	0.8106305805	0.8106305886	8.1×10^{-9}	J123	1.0640930	1.0640929688	1.0640929700	1.2×10^{-9}
J4	0.6861985	0.6861985109	0.6861985115	6.0×10^{-10}	J124	0.9409648	0.9409648534	0.9409648733	2.0×10^{-8}
J5	0.6520359	0.6520358928	0.6520359510	5.8×10^{-8}	J125	1.0000000	0.9999999999	1.0000000031	3.2×10^{-9}
J6	0.4461508	0.4461507712	0.4461507727	1.5×10^{-9}	J126	0.8023577	0.8023576521	0.8023576563	4.2×10^{-9}
J7	0.6057263	0.6057263191	0.6057263350	1.6×10^{-8}	J127	0.6718972	0.6718971603	0.6718971632	2.9×10^{-9}
J8	0.4772413	0.4772413037	0.4772413307	2.7×10^{-8}	J128	1.0096066	1.0096065872	1.0096066267	4.0×10^{-8}
J9	0.5000000	0.4999999999	0.5000000263	2.6×10^{-8}	J129	1.0522569	1.0522568555	1.0522568770	2.1×10^{-8}
J10	0.6938425	0.6938424637	0.6938424675	3.8×10^{-9}	A5	0.4353342	0.4353342463	0.4353342472	9.0×10^{-10}
J11	0.4684758	0.4684758078	0.4684758099	2.1×10^{-9}	A6	0.3003638	0.3003638383	0.3003638395	1.2×10^{-9}
J12	0.7261971	0.7261970814	0.7261970866	5.2×10^{-9}	A8	0.5916501	0.5916500663	0.5916500761	9.8×10^{-9}
J13	0.6926870	0.6926870296	0.6926870324	2.8×10^{-9}	A9	0.4652428	0.4652428455	0.4652428487	3.2×10^{-9}
J14	0.6317469	0.6317469144	0.6317469257	1.1×10^{-8}	A10	0.4158004	0.4158004469	0.4158004496	2.7×10^{-9}
J15	0.7808500	0.7808499742	0.7808499864	1.2×10^{-8}	A11	0.4561079	0.4561079332	0.4561079346	1.4×10^{-9}
J16	0.5674133	0.5674132998	0.5674133222	2.2×10^{-8}	A12	0.4877093	0.4877092778	0.4877092918	1.4×10^{-8}
J17	0.6379629	0.6379629132	0.6379629179	4.7×10^{-9}	A13	0.4252330	0.4252330452	0.4252330474	2.2×10^{-9}
J18	1.0130080	1.0130080047	1.0130080051	4.0×10^{-10}	A14	0.4759513	0.4758457381	–	–
J19	0.6742346	0.6742346141	0.6742346312	1.7×10^{-8}	A15	0.4496279	0.4496279430	0.4496279449	1.9×10^{-9}
J20	0.5932470	0.5932469972	0.5932470324	3.5×10^{-8}	A16	0.4571068	0.4571067811	0.4571067825	1.4×10^{-9}
J21	0.7017468	0.7017468424	0.7017468573	1.5×10^{-8}	A17	0.3754473	0.3754473439	0.3754473442	3.0×10^{-10}
J22	0.8156179	0.8156179275	0.8156179320	4.5×10^{-9}	A18	0.3843551	0.3843551338	0.3843551353	1.5×10^{-9}
J23	0.6090289	0.6090288648	0.6090288859	2.1×10^{-8}	A19	0.6226300	0.6226304433	0.6226304480	4.7×10^{-9}
J24	0.6942505	0.6942504595	0.6942504760	1.6×10^{-8}	A20	0.6022398	0.6022398259	0.6022398311	5.2×10^{-9}
J25	0.5147829	0.5147829042	0.5147829175	1.3×10^{-8}	A21	0.3260601	0.3260600674	0.3261777341	1.2×10^{-4}
J26	0.6402494	0.6402493792	0.6402493819	2.7×10^{-9}	A22	0.6234571	0.6234570835	0.6234570956	1.2×10^{-8}
J27	0.9642857	0.9642857142	0.9642857145	3.0×10^{-10}	A23	0.5460735	0.5460734856	0.5460734859	3.0×10^{-10}
J28	0.7500000	0.7499999999	0.7500000123	1.2×10^{-8}	A24	0.6047986	0.6047985728	0.6047985820	9.2×10^{-9}
J29	1.0245760	1.0245760143	1.0245760179	3.6×10^{-9}	A25	0.6033789	0.6033789232	0.6033789290	5.8×10^{-9}
J30	0.4363842	0.4363842240	0.4365201856	1.4×10^{-4}	A26	0.5275550	0.5275549713	0.5275550044	3.3×10^{-8}
J31	0.4772413	0.4772413037	0.4772413055	1.8×10^{-9}	A27	0.6483073	0.6483073083	0.6483073218	1.4×10^{-8}
J32	0.5901592	0.5901592141	0.5901592219	7.8×10^{-9}	A28	0.6403143	0.6403143403	0.6403143404	1.0×10^{-10}
J33	0.6150642	0.6150641795	0.6150641976	1.8×10^{-8}	A29	0.4920635	0.4920635075	0.4920635103	2.8×10^{-9}
J34	0.4492657	0.4492657413	0.4492657466	5.3×10^{-9}	A30	0.5698209	0.5698209396	0.5698209479	8.3×10^{-9}
J35	0.6932503	0.6932503121	0.6932503141	2.0×10^{-9}	A31	0.5738173	0.5738172956	0.5738172971	1.5×10^{-9}
J36	0.6706462	0.6706462343	0.6706462406	6.3×10^{-9}	A32	0.4135530	0.4135530286	0.4135530303	1.7×10^{-9}
J37	0.4864826	0.4864826234	0.4864826245	1.1×10^{-9}	A33	0.6226313	0.6226312693	0.6226312705	1.2×10^{-9}
J38	0.9295153	0.9295153311	0.9295153363	5.2×10^{-9}	A34	0.5350117	0.5350117257	0.5350117269	1.2×10^{-9}
J39	0.8264535	0.8264534944	0.8264534953	9.0×10^{-10}	A35	0.6249079	0.6249078573	0.6249078581	8.0×10^{-10}
J40	0.9208907	0.9208906711	0.9208906793	8.2×10^{-9}	A36	0.4388685	0.4388685039	0.4388685047	8.0×10^{-10}
J41	0.7596044	0.7596043867	0.7596043876	9.0×10^{-10}	A37	0.4868868	0.4868868202	0.4868868210	8.0×10^{-10}
J42	0.6722257	0.6722256558	0.6722256599	4.1×10^{-9}	A38	0.4699126	0.4699126549	0.4699126558	9.0×10^{-10}
J43	0.4685939	0.4685938833	0.4685938859	2.6×10^{-9}	A39	0.6172035	0.6172034826	0.6172035172	3.5×10^{-8}
J44	0.4144533	0.4144532761	0.4144533354	5.9×10^{-8}	A40	0.6078638	0.6078638262	0.6078638382	1.2×10^{-8}
J45	0.7705600	0.7705599645	0.7705599760	1.1×10^{-8}	A41	0.4785634	0.4785634179	0.4785634189	1.0×10^{-9}
J46	0.9716836	0.9716836271	0.9716836603	3.3×10^{-8}	A42	0.6198655	0.6198655036	0.6198655293	2.6×10^{-8}
J47	0.7644749	0.7644749219	0.7644749489	2.7×10^{-8}	A43	0.6107654	0.6107654035	0.6107654110	7.5×10^{-9}
J48	0.7516220	0.7516219860	0.7516256937	3.7×10^{-6}	A44	0.5364942	0.5364942157	0.5364942161	4.0×10^{-10}
J49	0.8156923	0.8156923336	0.8156923392	5.6×10^{-9}	A45	0.5372394	0.5372393811	0.5372393837	2.6×10^{-9}
J50	0.8555932	0.8555932318	0.8555932323	5.0×10^{-10}	A46	0.4590108	0.4590108470	0.4590108489	1.9×10^{-9}
J51	0.6750341	0.6750340746	0.6750341128	3.8×10^{-8}	A47	0.4608544	0.4608543948	–	–
J52	1.0999033	1.0999032950	1.0999033674	7.2×10^{-8}	A48	0.4631707	0.4631707404	0.4631707412	8.0×10^{-10}
J53	0.8093253	0.8093252883	0.8093253136	2.5×10^{-8}	A49	0.4666943	0.4666942750	0.4666942757	7.0×10^{-10}
J54	0.5194497	0.5194496955	0.5194497554	6.0×10^{-8}	A50	0.5182900	0.5182899837	0.5182899863	2.6×10^{-9}
J55	0.8283881	0.8283880914	0.8283881365	4.5×10^{-8}	A51	0.6607809	0.6607809402	0.6607809440	3.8×10^{-9}
J56	0.8192037	0.8192036987	0.8192036993	6.0×10^{-10}	A52	0.6218611	0.6218611548	0.6218611570	2.2×10^{-9}
J57	0.8609885	0.8609884906	0.8609885652	7.5×10^{-8}	A53	0.6386102	0.6386102120	0.6386102158	3.8×10^{-9}
J58	0.8814481	0.8814480649	0.8814480651	2.0×10^{-10}	A54	0.5936813	0.5936812986	0.5936813002	1.6×10^{-9}
J59	0.6379629	0.6379629132	0.6379629144	1.2×10^{-9}	A55	0.6213203	0.6213203435	0.6213203751	3.2×10^{-8}
J60	0.5922714	0.5922713726	0.5922713739	1.3×10^{-9}	A56	0.6893124	0.6893123503	0.6893123573	7.0×10^{-9}
J61	0.8175538	0.8175538005	0.8175538208	2.0×10^{-8}	A57	0.6603444	0.6603444166	0.6603444193	2.7×10^{-9}
J62	0.7500755	0.7500885544	0.7500900835	1.5×10^{-6}	A58	0.6488905	0.6488904507	0.6488904509	2.0×10^{-10}
J63	0.6078543	0.6078543740	0.6078543991	2.5×10^{-8}	A59	0.4488256	0.4488256417	0.4488256420	3.0×10^{-10}
J64	0.9167082	0.9167081796	0.9167082661	8.7×10^{-8}	A60	0.3940032	0.3940031953	0.3940032019	6.6×10^{-9}
J65	1.1110314	1.1110313940	1.1110314027	8.7×10^{-9}	A61	0.4019248	0.4019248079	0.4019248088	9.0×10^{-10}
J66	0.6186278	0.6186277769	0.6186277846	7.7×10^{-9}	A62	0.4065268	0.4065717137	–	–
J67	0.7212948	0.7212948125	0.7212948168	4.3×10^{-9}	A63	0.4894164	0.4894164301	0.4894164320	1.9×10^{-9}

ineq.	pub.	lower	upper	width	ineq.	pub.	lower	upper	width
J68	1.0178506	1.0178506212	1.0178506417	2.1×10^{-8}	A64	0.3900890	0.3900890509	–	–
J69	0.5081419	0.5081418854	0.5081418891	3.7×10^{-9}	A65	0.3688996	0.3688995748	0.3688995760	1.2×10^{-9}
J70	0.9538824	0.9538823982	0.9538823984	2.0×10^{-10}	A66	0.4877093	0.4877092778	0.4877092801	2.3×10^{-9}
J71	0.5822872	0.5822871551	0.5822872048	5.0×10^{-8}	A67	0.3990671	0.3990671029	0.3993386504	2.7×10^{-4}
J72	0.7878251	0.7878251303	0.7878251389	8.6×10^{-9}	A68	0.4025522	0.4028002584	–	–
J73	1.1223170	1.1223870369	1.1224104368	2.3×10^{-5}	A69	0.6096103	0.6096103211	0.6096103263	5.2×10^{-9}
J74	0.4348553	0.4348553293	0.4348553797	5.0×10^{-8}	A70	0.6052228	0.6052228085	0.6052228926	8.4×10^{-8}
J75	0.6270262	0.6270261912	0.6270261914	2.0×10^{-10}	A71	0.4490163	0.4490162651	0.4490162666	1.5×10^{-9}
J76	0.8291077	0.8291077045	0.8291077048	3.0×10^{-10}	A72	0.6962822	0.6962821908	0.6962821911	3.0×10^{-10}
J77	0.8406242	0.8406241785	0.8406241943	1.6×10^{-8}	A73	0.8831381	0.8831380531	0.8831380538	7.0×10^{-10}
J78	0.8937709	0.8937708980	0.8937709029	4.9×10^{-9}	A74	0.6890694	0.6890694094	0.6890694134	4.0×10^{-9}
J79	1.0156152	1.0156152324	1.0156152363	3.9×10^{-9}	A75	0.6051510	0.6051510348	0.6051510359	1.1×10^{-9}
J80	0.9051257	0.9051256536	0.9051257015	4.8×10^{-8}	A76	0.4898631	0.4898630974	0.4898630983	9.0×10^{-10}
J81	0.5995871	0.5995871500	0.5995871635	1.4×10^{-8}	A77	0.6655582	0.6655581944	0.6655581952	8.0×10^{-10}
J82	0.7308078	0.7308078152	0.7308078161	9.0×10^{-10}	A78	0.8927018	0.8927017674	0.8927017677	3.0×10^{-10}
J83	0.6691786	0.6691785966	0.6691786001	3.5×10^{-9}	A79	0.6243153	0.6243152878	0.6243152908	3.0×10^{-9}
J84	1.0669560	1.0669560027	1.0669560063	3.6×10^{-9}	A80	0.3769863	0.3769863185	–	–
J85	0.9762793	0.9762792520	0.9762792708	1.9×10^{-8}	A81	0.6690099	0.6690099536	0.6690099556	2.0×10^{-9}
J86	0.7558762	0.7558761808	0.7558762162	3.5×10^{-8}	A82	0.4708838	0.4709112502	–	–
J87	0.6848726	0.6848725982	0.6848726370	3.9×10^{-8}	A83	0.6961664	0.6961664112	0.6961664125	1.3×10^{-9}
J88	0.6159879	0.6159878918	0.6159879144	2.3×10^{-8}	A84	0.6352087	0.6352044080	–	–
J89	1.0035502	1.0035501556	1.0035502426	8.7×10^{-8}	A85	0.6411408	0.6411408410	0.6411408436	2.6×10^{-9}
J90	0.8398157	0.8398156462	0.8398156531	6.9×10^{-9}	A86	0.8004425	0.8004425430	0.8004425460	3.0×10^{-9}
J91	1.2992769	1.2992769151	1.2992769352	2.0×10^{-8}	A87	0.7562471	0.7562470649	0.7562470699	5.0×10^{-9}
J92	1.0648162	1.0648161983	1.0648162008	2.5×10^{-9}	A88	0.4142136	0.4142135623	0.4142136010	3.9×10^{-8}
J93	0.9627261	0.9627261088	0.9627261240	1.5×10^{-8}	A89	0.3035637	0.3039223061	–	–
J94	0.5944550	0.5944549695	0.5944549807	1.1×10^{-8}	AII1	0.6055543	0.6055542992	0.6055543099	1.1×10^{-8}
J95	1.0014114	1.0014114009	1.0014114251	2.4×10^{-8}	AII2	0.5000000	0.4999999999	0.5000000250	2.5×10^{-8}
J96	0.9416515	0.9416514703	0.9416514979	2.8×10^{-8}	AS1	0.5412415	0.5412414523	0.5412414553	3.0×10^{-9}
J97	1.1583626	1.1583625533	1.1583625755	2.2×10^{-8}	AS2	0.8784928	0.8784927944	0.8784927998	5.4×10^{-9}
J98	1.1496911	1.1496910757	1.1496911407	6.5×10^{-8}					

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